Seat No.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 | **Academic Year:** 2024

Subject: 031001122 Chemistry II | **Section:** 15-18

Date: 7 March 2025 | **Time:** 08.00 – 10.00

Name: ______________________ ID: ______________ Class: ______

Directions: The test is designed to measure your comprehension. The test is divided into 2 parts. There will be 8 pages (including this page) and they are worth 40 points.

- **This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.**
- Answer the questions on this test paper. วิทยาลัยเทคโนโลยีอุตสาหกรรม
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are **NOT** allowed in the examination.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:		Exam
CLO 1	Comprehend colligative properties and role of solution concentration in the properties.	Parts 1-2
CLO 2	Calculate changes in colligative properties for non-electrolyte and electrolyte solutes.	Parts 1-2
CLO 3	Understand and be able to differentiate an empirical formula and an molecular formula.	Parts 1-2
CLO 4	Define a limiting reagent and determine the limiting reagent in the reaction by calculation.	Parts 1-2
CLO 5	Learn definition and calculation related to %yield.	Parts 1-2
CLO 6	Explain periodic trend in detail.	Parts 1-2

Instructor: Assoc. Prof. Dr. Nathjanan Jongkon and Asst. Prof. Dr. Sunanta Chuayprakong

	Part 1	Part 2	Total
score			

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Gas constant or R = 0.082 atm•L/mol•K

Relative atomic mass of H =1, He =4, C =12, N =14, O =16, F =19, Na =23, S =32, Cl =35.5, Ar =40, K =39, Cu = 64, Zn = 65.5, Ag = 108, P = 30, Br = 80

Boiling Point Elevation Constants (K_b) and Freezing Point Depression Constants (K_f) for Some Solvents

Solvent	Boiling Point (°C)	K_b (°C/m)	Freezing Point (°C)	K_f (°C/m)
Water	100	0.51	0	1.86
Benzene	80.1	2.53	5.5	5.12
Acetic acid	118.1	3.07	16.6	3.90
Nitroenzene	210.9	5.24	5.7	7.00
Phenol	182	3.56	43	7.40
Camphor	207.4	5.61	178.4	40.0

Part 1. Multiple Choice (9 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

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1. Which one is correct?

   a. Semipermeable membrane allows solute to pass through it

   b. Colligative properties depend on the concentration of solute in the solution

   c. Colligative properties are the chemical properties of solution

   d. Colligative properties depend on type of solute structures

2. Each of the following solutions is added to equal amounts of water. Which solution will result in the greatest amount of boiling point elevation?

   a. 1.5 mol of $CaCl_2$ c. 1 mol of NaCl

   b. 2.5 mol of $KClO_4$ d. 2 mol of Na_2SO_4

3. Which statement is true?

   a. Formula which gives simplest whole number ratio of atoms is molecular formula.

   b. Molecular formula are always different from empirical formula.

   c. Empirical formula and molecular formula can be the same in certain cases.

   d. We don't need to know the molar mass of the compound in order to obtain the molecular formula.

4. The empirical formula of a hydrocarbon is CH_3 and its molar mass is 30 g/mol. What is the molecular formula?

   a. CH_2 b. C_2H_6 c. C_3H_9 d. C_4H_{10}

5. The molecular formula for vitamin C is $C_6H_8O_6$. What is the empirical formula mass?

   a. 22 b. 44 c. 88 d. 176

6. What are the coefficients respectively when the equation?

   $\_\_PH_3 + \_\_O_2 \rightarrow \_\_P_2O_5 + \_\_H_2O$ is balanced?

   a. 2, 2, 1, 3 b. 2, 1, 3, 4 c. 2, 3, 1, 2 d. 2, 4, 1, 3

7. Which one is incorrect when the equation?

   $C_4H_8\,(g) + O_2\,(g) \longrightarrow CO_2\,(g) + H_2O\,(g)$ is balanced?

   a. The reaction absorbs heat
   b. The coefficients will be 1, 6, 4, 4 respectively
   c. The summation of coefficients will be 15
   d. The maximum yield of water is 72 g

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8. The actual yield of a reaction is 30 g and the theoretical yield is 40 g. What is the percentage yield?

   a. 50 % b. 75 % c. 85 % d. 100 %

9. Which elements in the same period would have the strongest metallic character?

   a. Alkali b. Chalcogen c. Halogen d. Nobel gas

Part 2 Read the construction carefully and complete the information required. Students must provide calculation detail if CALCULATION is noted for any questions. Not following these directions could affect your score. (31 points)

1. Read the following sentences carefully and state whether they are **true** (T) or **false** (F). (5 points)

.... Electronegativity increases when atomic number decreases.

.... Atomic radius increases from the left to the right across the periodic table.

.... The first ionization energy is always lowest when compared to the second and the third ionization energy of the same element.

.... The reactant that controls the amount of product obtained is called the limiting reactant.

.... Theoretical yield is how much product you actually get from a reaction.

<u>Colligative properties</u>

2. **Calculate and compare** boiling points of the solutions **a** and **b** when water is the solvent for these two solutions. (Calculation, 3 points)
   a. 1 m KBr (ionizable, electrolyte solute)
   b. 1 m $C_{12}H_{22}O_{11}$ (non-electrolyte solute)

3. What is the vapor pressure of an aqueous solution that has a solute mole fraction of 0.4? The vapor pressure of water is 190 mmHg at 25 °C. (Calculation, 2 points)

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4. Automotive antifreeze consists of ethylene glycol, $C_2H_6O_2$, a nonvolatile nonelectrolyte. Calculate the freezing point of a 10 % by mass solution of ethylene glycol in water. (Calculation, 3 points)

<u>Empirical Formula and molecular formula</u>

5. The molar mass of a compound is 92 g/mol. Analysis of a sample of the compound indicates that it contains 0.606 g N and 1.390 g O. Find its molecular formula.

   (Calculation, 3 points)

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6. A compound has the empirical formula C_2H_4O, and its formula mass is 88 g. What is its molecular formula? (Calculation, 2 points)

7. A compound of carbon and hydrogen contains 92.3% C and has a molar mass of 78 g/mol. Determine the empirical and molecular formulas of the compound. (Calculation, 3 points)

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Limiting Agent and Yield percentage

8. Given the following reaction. (Calculation, 4 points)

$$C_3H_8 \quad + \quad O_2 \quad \text{------->} \quad CO_2 \quad + \quad H_2O$$

Balance the chemical equation

.... C_3H_8 (g) +....... O_2 (g) -------> .... CO_2 (g) + ...H_2O (g)

a) What is the limiting reagent, if you start with 22 g of C_3H_8 and 16 g of O_2?

b) How many liters of carbon dioxide can be produced from the reaction at STP?

c) Determine the amount in grams of excess reactant left.

9. Given the following reaction. (Calculation, 3 points)

$$CH_3OH\ (l) \quad + \quad CO\ (g) \quad \text{------->} \quad CH_3COOH\ (l)$$

If 24 g of CH_3OH reacts with 14 g of CO,

a. which is the limiting reagent?

b. what is the amount in grams (theoretical yield) of CH_3COOH?

c. If the actual yield of CH_3COOH is 11.25 g, what is the percentage yield?

10. $CO\ (g) + H_2\ (g)$ -------> $CH_3OH\ (l)$

If 6 g of hydrogen gas is reacted with 44.8 L of carbon monoxide at STP. What mass of CH_3OH is produced? (Calculation, 3 points)

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